Mini Project: Cohort 21: Forest Fire Ignition Risk

Date:07/03/2026

In today's meeting, our group had a discussion regarding the progress made by the data team in preparing the dataset. During the meeting, the data team explained the steps they followed to improve the dataset and make it more suitable for further analysis and model development.

For this project, the dataset was obtained from Kaggle, an online platform that provides open-source datasets for research and machine learning projects. The team selected a suitable dataset related to environmental factors that influence forest fire ignition.

After loading the dataset, the data team worked on feature engineering to enhance the dataset and extract more meaningful information from the existing variables. New features such as Temperature × Humidity and Temperature / Humidity ratio were created to better understand how the interaction between temperature and humidity affects fire risk conditions.

The team also calculated the Heat Index, which represents the perceived temperature when humidity is considered along with the actual temperature. This helps in identifying environmental conditions that may increase the chances of forest fire ignition.

In addition, new columns such as Classes and Ignition Risk Classes were added to the dataset. The ignition risk was categorized into three levels: Low, Moderate, and High, which will help in building classification models and interpreting the results more clearly.

Processed Dataset:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	day	month	year	Temperatu	RH	Ws	Rain	FFMC	DMC	DC	ISI	BUI	FWI	Classes	Temp_Hur	Temp_Hur	Heat_Inde	Fire_Risk	Fire_Risk_I	Fire_Risk_Encoded	
2	1	6	2012	-0.87065	-0.34067	0.89328	-0.38166	-0.84793	-0.91212	-0.8794	-0.83036	-0.93601	-0.88016		-0.80596	-0.25709	-1.038	Low	0	0	
3	2	6	2012	-0.87065	-0.07036	-0.88888	0.26864	-0.93871	-0.85552	-0.8794	-0.90272	-0.9008	-0.89363		-0.5013	-0.4031	-0.98532	Low	0	0	
4	3	6	2012	-1.69925	1.34879	2.319	6.17136	-2.1468	-0.98489	-0.88991	-1.07157	-0.98531	-0.93403		0.45211	-1.09602	-1.63344	Low	0	0	
5	4	6	2012	-1.97545	1.82184	-0.88888	0.86892	-3.43869	-1.08192	-0.89412	-1.14394	-1.05574	-0.9475		0.69637	-1.25433	-1.84947	Low	0	0	
6	5	6	2012	-1.42305	1.0109	0.18042	-0.38166	-0.91078	-0.94446	-0.74065	-0.85448	-0.9008	-0.88016		0.3129	-0.94908	-1.39106	Low	0	0	
7	6	6	2012	-0.31825	0.33511	-0.53245	-0.38166	0.33223	-0.71806	-0.57247	-0.39617	-0.68247	-0.61081		0.30765	-0.45878	-0.28985	Low	0	0	
8	7	6	2012	0.23414	-0.54341	-0.88888	-0.38166	0.72329	-0.38655	-0.39798	0.39984	-0.40781	0.02217		-0.46715	0.19077	0.1554	Moderate	1	1	
9	8	6	2012	-0.59445	0.74058	-0.17602	-0.38166	0.61156	-0.20866	-0.234	0.20687	-0.2247	0.0087		0.60444	-0.68515	-0.51906	Moderate	1	1	
10	9	6	2012	-1.97545	1.75426	-0.88888	-0.28161	-1.74178	-0.54826	-0.22349	-1.04745	-0.43598	-0.90709		0.63071	-1.24036	-1.86264	Low	0	0	
11	10	6	2012	-1.14685	1.14605	-1.24531	-0.38166	-0.32419	-0.41889	-0.06582	-0.83036	-0.28808	-0.82629		0.66222	-0.93253	-1.05649	Low	0	0	
12	11	6	2012	-0.31825	0.19995	-0.53245	-0.38166	0.46491	-0.17632	0.10236	-0.17907	-0.06272	-0.19331		0.14481	-0.39647	-0.31619	Moderate	1	1	
13	12	6	2012	-1.69925	1.28121	1.24971	-0.38166	0.43	-0.07121	0.25162	0.0139	0.07109	0.0087		0.38382	-1.07889	-1.64661	Moderate	1	1	
14	13	6	2012	-1.42305	1.48394	1.96257	0.21862	-1.94429	-0.64529	-0.68179	-1.02333	-0.7036	-0.92056		0.8093	-1.07696	-1.29887	Low	0	0	
15	14	6	2012	-0.59445	1.07847	1.60614	-0.13154	-1.3158	-0.81509	-0.8752	-0.90272	-0.86558	-0.89363		0.99841	-0.80044	-0.45321	Low	0	0	
16	15	6	2012	-1.14685	1.21363	0.53685	1.16906	-1.98619	-0.94446	-0.88361	-1.04745	-0.96418	-0.93403		0.73576	-0.95192	-1.04332	Low	0	0	
17	16	6	2012	-0.87065	1.82184	-0.88888	-0.0315	-2.91495	-1.04958	-0.8794	-1.14394	-1.02052	-0.9475		1.63138	-1.05764	-0.61657	Low	0	0	
18	17	6	2012	-0.59445	1.82184	0.18042	-0.08152	-2.83116	-1.09809	-0.8752	-1.14394	-1.06278	-0.9475		1.86514	-1.00847	-0.30834	Low	0	0	
19	18	6	2012	-0.31825	1.07847	-0.53245	-0.23159	-1.46245	-1.03341	-0.87099	-0.97508	-1.00644	-0.92056		1.20327	-0.74433	-0.14498	Low	0	0	
20	19	6	2012	-0.31825	-0.47583	0.18042	-0.33163	0.14369	-0.82318	-0.70281	-0.5409	-0.8022	-0.75895		-0.66939	-0.01699	-0.44789	Low	0	0	
21	20	6	2012	-0.59445	1.21363	0.18042	-0.18156	-1.25994	-0.91212	-0.46946	-0.92684	-0.81629	-0.89363		1.156	-0.84252	-0.42687	Low	0	0	
22	21	6	2012	-0.59445	1.07847	-0.53245	-0.38166	0.2205	-0.67763	-0.37485	-0.51678	-0.58388	-0.65121		0.99841	-0.80044	-0.45321	Low	0	0	
23	22	6	2012	-0.31825	0.33511	0.53685	-0.33163	0.08782	-0.62103	-0.20877	-0.56502	-0.49232	-0.63774		0.30765	-0.45878	-0.28985	Low	0	0	
24	23	6	2012	-0.04206	-0.00278	0.89328	-0.33163	0.24844	-0.52401	-0.03639	-0.34793	-0.36555	-0.43573		0.06339	-0.2249	-0.04747	Low	0	0	
25	24	6	2012	-0.04206	0.26753	0.53685	-0.38166	0.56268	-0.28144	0.1339	0.20687	-0.1261	0.06257		0.39958	-0.36179	0.00521	Moderate	1	1	

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
26	25	6	2012	-0.31825	0.13238	-0.17602	-0.38166	0.61854	-0.03886	0.30208	0.23099	0.11335	0.18378		0.06339	-0.36386	-0.32936	Moderate	1	1	
27	26	6	2012	-0.31825	0.13238	0.89328	-0.38166	0.62553	0.25222	0.47026	0.47221	0.34576	0.48007		0.06339	-0.36386	-0.32936	Moderate	1	1	
28	27	6	2012	0.51034	-0.61099	0.89328	-0.38166	0.77916	0.55948	0.64896	1.07524	0.64155	1.07264		-0.41462	0.3238	0.45045	Severe	2	2	
29	28	6	2012	-0.04206	-0.47583	-0.53245	-0.38166	0.78614	0.87482	0.82134	0.6893	0.91622	0.9245		-0.52493	0.06258	-0.13966	Moderate	1	1	
30	29	6	2012	-0.04206	-1.01646	-0.88888	-0.23159	0.14369	0.30073	0.73515	-0.61326	0.5007	-0.42226		-1.19731	0.49597	-0.24502	Low	0	0	
31	30	6	2012	0.23414	-0.81372	-0.53245	-0.38166	0.75821	0.66459	0.91174	0.59281	0.81762	0.78982		-0.81384	0.40472	0.10272	Moderate	1	1	
32	1	7	2012	-0.87065	0.40269	1.24971	0.11857	-1.25295	-0.98489	-0.85838	-0.8786	-0.97123	-0.89363		0.03187	-0.61727	-0.89313	Low	0	0	
33	2	7	2012	-1.42305	0.87574	1.24971	0.21862	-1.54625	-0.99298	-0.86469	-0.95096	-0.97827	-0.90709		0.17108	-0.90816	-1.4174	Low	0	0	
34	3	7	2012	-0.04206	0.94332	1.60614	-0.0315	-1.02949	-0.97681	-0.84576	-0.83036	-0.96418	-0.88016		1.24004	-0.64098	0.1369	Low	0	0	
35	4	7	2012	0.23414	1.07847	0.53685	-0.38166	0.15765	-0.81509	-0.65025	-0.49265	-0.77403	-0.71855		1.613	-0.63212	0.47147	Low	0	0	
36	5	7	2012	0.23414	0.26753	-0.53245	-0.38166	0.56268	-0.57252	-0.45264	0.0139	-0.53458	-0.28759		0.57292	-0.29548	0.31343	Low	0	0	
37	6	7	2012	-0.04206	0.0648	-0.53245	-0.38166	0.63949	-0.30569	-0.26133	0.20687	-0.29513	-0.0317		0.14744	-0.26075	-0.0343	Moderate	1	1	
38	7	7	2012	0.78654	0.13238	0.89328	-0.28161	0.15067	-0.40272	-0.18985	-0.46853	-0.3233	-0.51653		0.73576	-0.09034	0.90354	Low	0	0	
39	8	7	2012	0.23414	0.40269	1.24971	-0.38166	0.54173	-0.17632	0.00776	0.30336	-0.09089	0.12991		0.74627	-0.35984	0.33977	Moderate	1	1	
40	9	7	2012	-0.04206	0.40269	-0.53245	0.31866	-0.78508	-0.56444	-0.84576	-0.8786	-0.6543	-0.86669		0.56767	-0.4242	0.03154	Low	0	0	
41	10	7	2012	0.23414	0.47027	-0.88888	-0.0315	-0.78508	-0.70189	-0.84366	-0.8786	-0.76699	-0.88016		0.83294	-0.39062	0.35294	Low	0	0	
42	11	7	2012	0.23414	0.94332	-0.53245	-0.38166	0.22749	-0.53209	-0.64605	-0.51678	-0.605	-0.65121		1.43965	-0.58339	0.44513	Low	0	0	
43	12	7	2012	-0.31825	0.87574	-0.88888	-0.33163	-0.19151	-0.54826	-0.45684	-0.78211	-0.52754	-0.82629		0.95901	-0.67476	-0.18449	Low	0	0	
44	13	7	2012	0.51034	1.28121	-0.17602	-0.38166	0.27637	-0.40272	-0.25713	-0.42029	-0.35147	-0.4896		2.08576	-0.64666	0.8192	Low	0	0	
45	14	7	2012	0.51034	-0.07036	-0.88888	-0.08152	-0.2753	-0.55635	-0.55775	-0.80623	-0.58388	-0.83976		0.29977	-0.04438	0.55581	Low	0	0	
46	15	7	2012	-0.59445	1.21363	1.24971	-0.18156	-1.19709	-0.76658	-0.68179	-0.8786	-0.75994	-0.88016		1.156	-0.84252	-0.42687	Low	0	0	
47	16	7	2012	-1.14685	0.94332	1.96257	-0.38166	-0.36609	-0.62103	-0.50309	-0.97508	-0.59092	-0.89363		0.4416	-0.87131	-1.096	Low	0	0	
48	17	7	2012	-0.87065	0.53785	-0.53245	-0.38166	0.3462	-0.42698	-0.3223	-0.37205	-0.39372	-0.46266		0.18421	-0.67059	-0.86679	Low	0	0	
49	18	7	2012	-0.31825	0.40269	-0.53245	-0.38166	0.52776	-0.20866	-0.13309	-0.03435	-0.1754	-0.13944		0.38907	-0.48855	-0.27668	Moderate	1	1	
50	19	7	2012	0.78654	-0.20552	0.53685	-0.38166	0.71631	-0.21675	0.07083	0.71342	0.10631	0.52047		0.27613	0.11248	0.8377	Moderate	1	1	

In addition, the dataset was enhanced by adding Classes and Ignition Risk Classes. The ignition risk was categorized into three levels: Low, Moderate, and High, which will help in better understanding the fire risk and will also be useful while building machine learning models.

After creating these new features, the team performed feature scaling so that all the values in the dataset are within a similar range. This step helps improve the performance of machine learning models and ensures that no single feature dominates due to larger numerical values.

Finally, the newly created features and the scaled values were added back to the dataset, making it more informative and ready for the next stages of the project such as analysis and model building.

Code Snippets:

```
[8]
import pandas as pd
import numpy as np

[21]
df = pd.read_csv("algerian_forest_fire_with_features.csv")
df.head()
...

[22]
df.shape
... (243, 17)

[23]
df.columns
... Index(['day', 'month', 'year', 'Temperature', 'RH', 'Ws', 'Rain', 'FFMC',
       'DMC', 'DC', 'ISI', 'BUI', 'FWI', 'Classes', 'Temp_Humidity',
       'Temp_Humidity_Ratio', 'Heat_Index'],
      dtype='object')

[24]
df.info()
... <class 'pandas.core.frame.DataFrame'>
```

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 243 entries, 0 to 242
Data columns (total 17 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   day                  243 non-null    int64
 1   month                243 non-null    int64
 2   year                 243 non-null    int64
 3   Temperature          243 non-null    int64
 4   RH                   243 non-null    int64
 5   Ws                   243 non-null    int64
 6   Rain                 243 non-null    float64
 7   FFMC                 243 non-null    float64
 8   DMC                  243 non-null    float64
 9   DC                   243 non-null    float64
 10  ISI                  243 non-null    float64
 11  BUI                  243 non-null    float64
 12  FWI                  243 non-null    float64
 13  Classes              243 non-null    int64
 14  Temp_Humidity        243 non-null    int64
 15  Temp_Humidity_Ratio  243 non-null    float64
 16  Heat_Index           243 non-null    float64
dtypes: float64(9), int64(8)
memory usage: 32.4 KB

[25]
df.columns = df.columns.str.strip()
df.columns
... Index(['day', 'month', 'year', 'Temperature', 'RH', 'Ws', 'Rain', 'FFMC',
       'DMC', 'DC', 'ISI', 'BUI', 'FWI', 'Classes', 'Temp_Humidity',
       'Temp_Humidity_Ratio', 'Heat_Index'],
```

```
[25]
df.columns = df.columns.str.strip()
df.columns
... Index(['day', 'month', 'year', 'Temperature', 'RH', 'Ws',
       'DMC', 'DC', 'ISI', 'BUI', 'FWI', 'Classes', 'Temp
       'Temp_Humidity_Ratio', 'Heat_Index'],
      dtype='object')

[29]
df['Classes'].unique()
... array([0, 1], dtype=int64)

[31]
df.isnull().sum()
... day            0
month          0
year           0
Temperature    0
RH             0
Ws             0
Rain           0
FFMC           0
DMC            0
DC             0
ISI            0
BUI            0
FWI            0
Classes        0
```

```
[32]
df = df[df['Classes'] != 'Classes']

[33]
df['Classes'].unique()
... array([0, 1], dtype=int64)

[34]
df = df.dropna()

[35]
df.isnull().sum()
... day              0
month            0
year             0
Temperature      0
RH               0
Ws               0
Rain             0
FFMC             0
DMC              0
DC               0
ISI              0
BUI              0
FWI              0
Classes          0
Temp_Humidity    0
```

```
[36]
df['Classes'] = df['Classes'].map({'not fire':0, 'fire':1})
```

```
print(df['Classes'].unique())
print(df['Classes'].value_counts())
```

```
[nan]
Series([], Name: count, dtype: int64)
```

```
[40]
for column in df.columns:
    if column != 'Classes':
        df[column] = pd.to_numeric(df[column])
```

```
[41]
df.info()
df.shape
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 243 entries, 0 to 242
Data columns (total 17 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   day             243 non-null    int64
 1   month           243 non-null    int64
 2   year            243 non-null    int64
 3   Temperature     243 non-null    int64
```

```
df.to_csv("algerian_forest_fire_with_features.csv", index=False)
```

```
import numpy as np

# to make sure FWI is numeric
df['FWI'] = pd.to_numeric(df['FWI'], errors='coerce')

conditions = [
    (df['FWI'] < 5),
    (df['FWI'] >= 5) & (df['FWI'] < 15),
    (df['FWI'] >= 15)
]

risk_levels = ['Low', 'Moderate', 'Severe']

df['Fire_Risk'] = np.select(conditions, risk_levels, default='Low')
```

```
df['Fire_Risk_Label'] = df['Fire_Risk'].map({
    'Low':0,
    'Moderate':1,
    'Severe':2
})
```

```
print(df[['FWI','Fire_Risk','Fire_Risk_Label']])
```

Overall, the meeting helped all team members understand the work done by the data team and the improvements made to the dataset.